

Analyzing Mental Health Conversations with ML + SQL

Tony Wu
03/30/2025

Business Problem

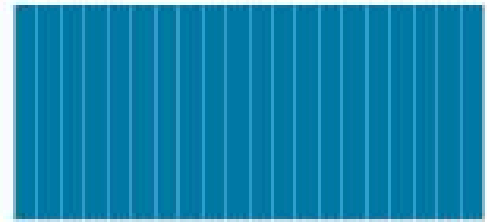
- Therapy sessions contain emotional and diagnostic signals, but they are difficult to measure or analyze at scale.
- Therapists need an objective tool to support their judgment and reduce over-reliance on personal interpretation.
- Insurance companies need a clear and fair way to evaluate how severe a case is, and how much coverage to provide.
- There is currently no tool that analyzes therapy dialogue for both clinical and insurance-related decisions.

If we can build a prediction model using machine learning, therapists could better track a patient's condition objectively over time, and insurance companies could make faster, clearer decisions about eligibility and payment.

Dataset and Approach

Dataset: “Sentiment Analysis for Mental Health” (Kaggle, 51,000+ sentences)

- Each sentence is labeled with a diagnosis category (e.g., Depression, Suicidal, Anxiety)

unique_id	Statement	Mental Health Status	
 053.0k	51074 unique values	Normal	31%
		Depression	29%
		Other (21288)	40%
7035	Today is the last day of being 27. Tbh I always thought I will die way before 28. Life is not great ...	Depression	
7036	I have a close friend who has suffered from periods of depression for years. she is been doing well ...	Depression	
7037	So i do not take it for BP, i take it for PTSD, OCD and migraines and so far it aplears to be liftin...	Depression	

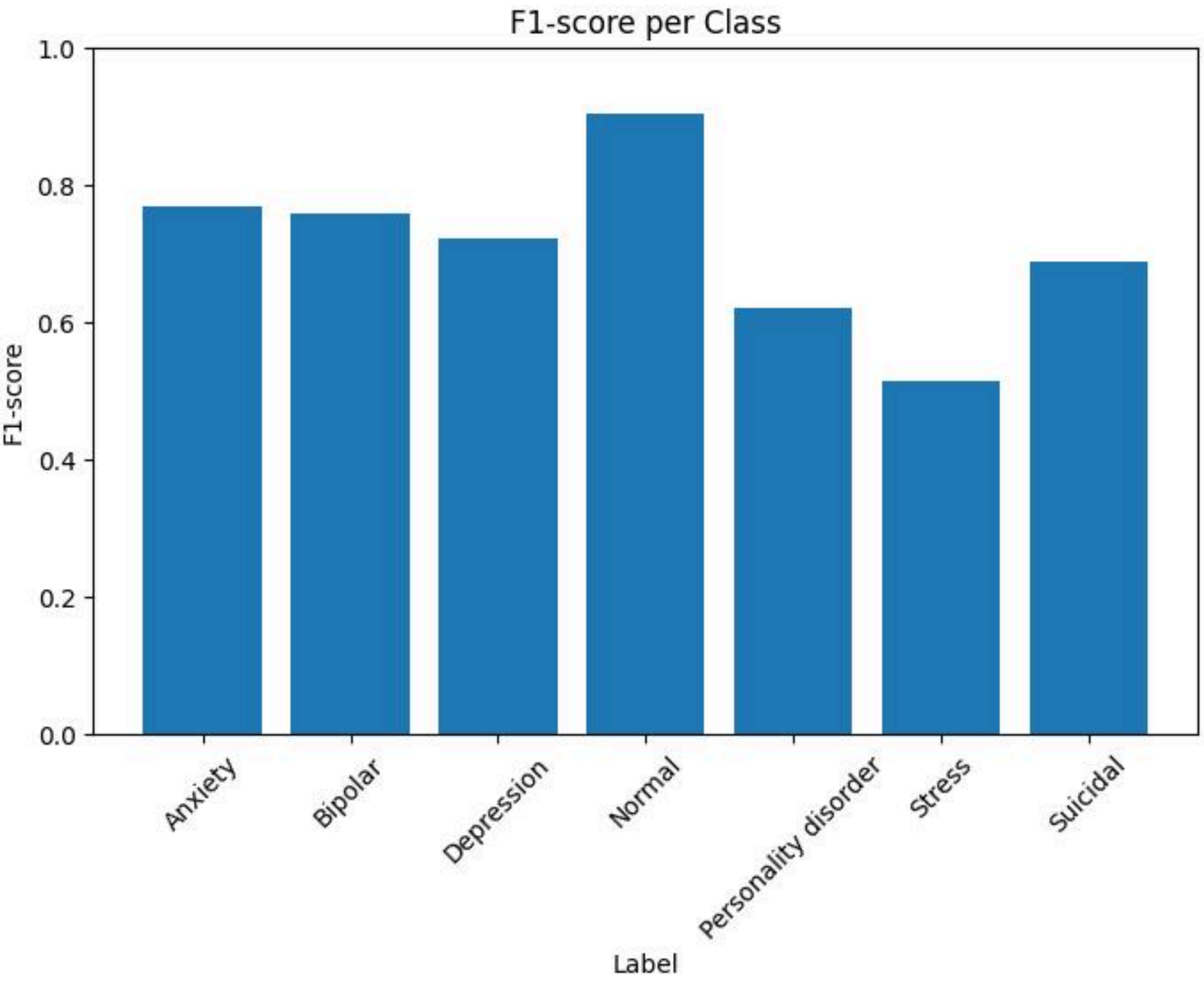
Preprocessing:

1. Cleaned the data by removing empty or invalid entries
2. Used TF-IDF to convert each sentence into numbers, based on word patterns

Model: Trained a Logistic Regression classifier to predict diagnosis (used as a baseline model for interpretability and speed)

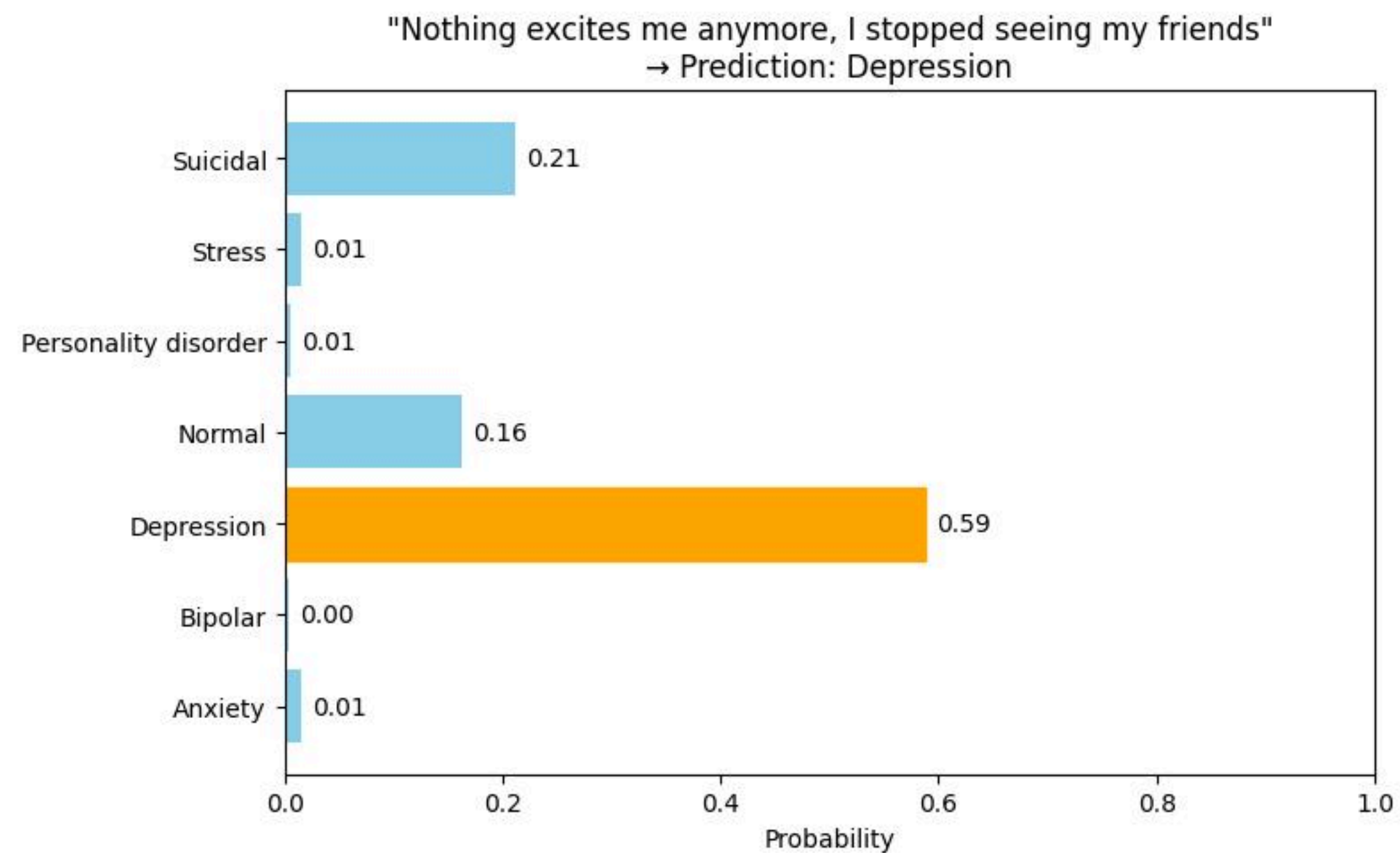
Model Performance

•Overall accuracy: 77%

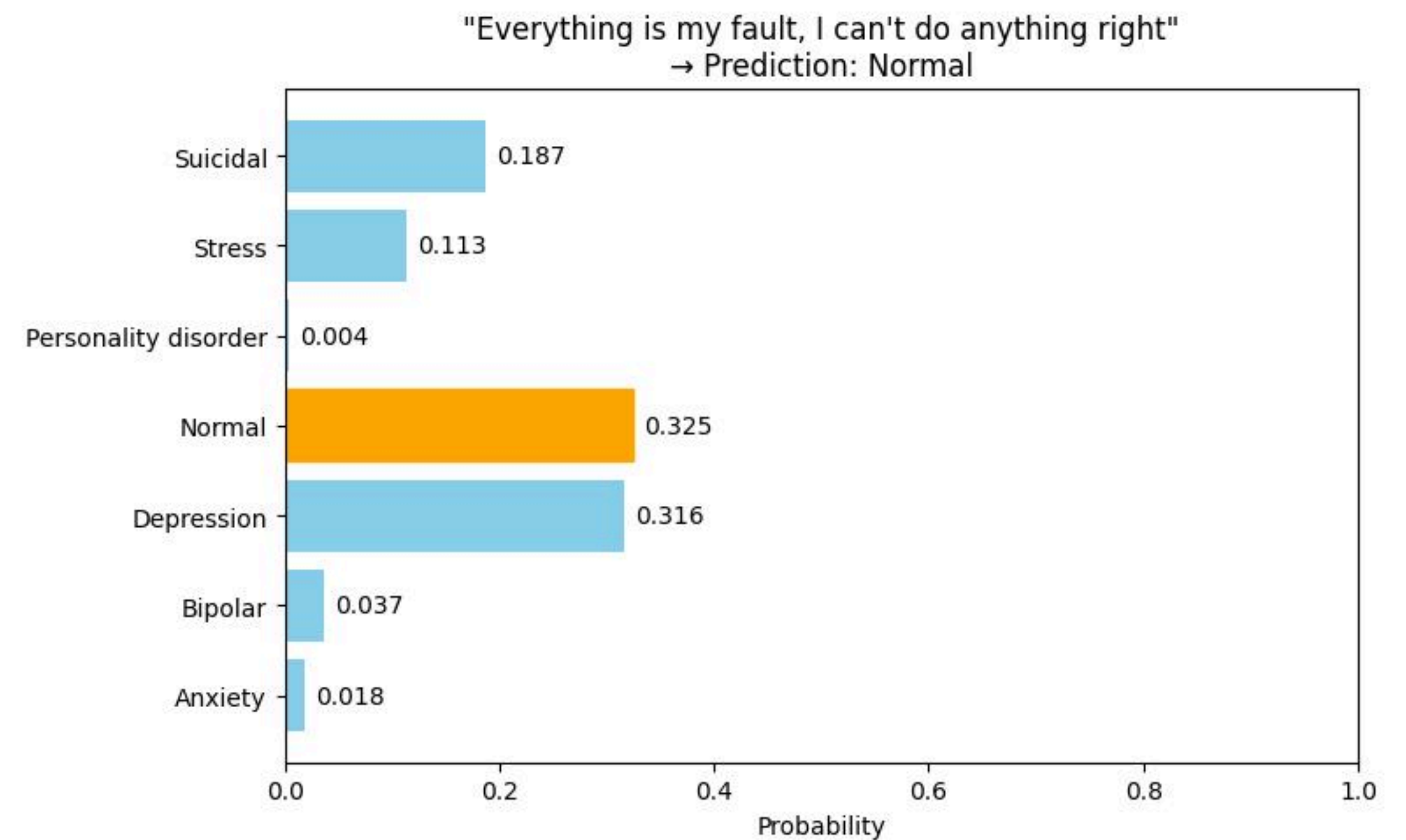


Testing on Real Dialogue

Example input: "Nothing excites me anymore, I stopped seeing my friends"



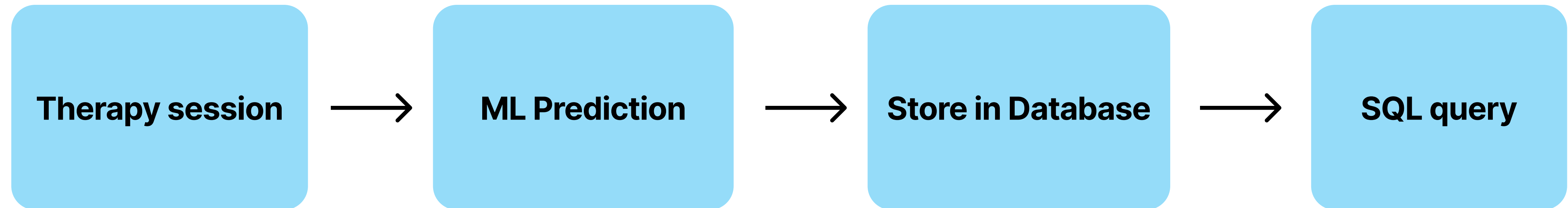
Example input: "Everything is my fault, I can't do anything right"



Limitations and Future Directions

- Some categories (Normal) have much more data than others, so the model gets used to predicting the common ones.
 - The model can't fully understand subtle or complex emotions in a sentence.
 - Logistic Regression lacks context awareness.
-
- Future models can use BERT to better understand emotional meaning in context.
 - Improving data labeling quality (e.g., with clinical expert input) can help the model learn more accurately.

Possible SQL Query



E.g. Find potentially high-risk sentences

```
SELECT sentence, prob_suicidal  
FROM therapy_predictions  
WHERE prob_suicidal > 0.5  
ORDER BY prob_suicidal DESC;
```

E.g. Count the number of high-risk sentences in each session

```
SELECT session_id, COUNT(*) AS  
high_risk_count  
FROM therapy_predictions  
WHERE prob_suicidal > 0.5  
GROUP BY session_id;
```

Three Questions to Think About

- 1. How can we build models that understand emotions better, not just keywords?**
- 2. What are the risks of using AI to guess someone's mental health?**
- 3. How should we balance catching risky cases and not overreacting?**